

Tang Song Soundscape – Project Report

1. Research Question

This project maps the “soundscapes” of classical Chinese literature. By analyzing 300 Tang poems and 300 Song lyrics, I explore how poetic sounds changed across space and time.

Main RQ: How does the spatial distribution of auditory imagery (soundscapes) in Chinese classical poetry shift between the Tang and Song dynasties?

Sub RQ 1: Based on the density of auditory descriptions, what are the characteristics of the geographic locations most frequently associated with sound across different periods?

Sub RQ 2: What are the differences in categories of sound events at different times or in different locations?

I hypothesize that the transition from Tang to Song reflects a clear spatial shift in soundscapes—from expansive, frontier-oriented sounds (military horns, desert winds) clustered along northern borders, to more urbanized, domestic sounds concentrated in the southern commercial hubs (Jiangnan region).

2. Data Sources

2.1 Data Acquisition and Preprocessing

The core data foundation of my project is *Three Hundred Tang Poems* and *Three Hundred Song Ci (chinese-poetry, n.d.)*. Very luckily, I found an existing JSON format database on GitHub. After getting the poetry texts, I first did a Traditional-to-Simplified Chinese conversion for *Three Hundred Tang Poems*, and then initially determined the dimensions (i.e., data columns) needed for my database.

The first one I built was the poetry ontology database, containing the dimensions: Poem ID, Dynasty, Genre, as well as the author and title.

2.2 Database Construction and Validation of Sound Events

Next, I built a Sound Entry Level database, mainly including the following dimensions:

- **The sound event itself:** Including Original Phrase, Sound Source, Context, Window (time/space window), and Category.
- **Perception Modality:** This item is crucial. Because sometimes the poet explicitly wrote what they “heard” (with a verb); sometimes they only described bird songs or rain sounds without a “listening” verb, but auditory presence can be inferred; and sometimes the poet only wrote “strong wind blowing,” which inevitably comes with

sound even if not directly written. Therefore, I divided the perception modality into these three categories.

- **Perception Reality:** That is, whether this sound was actually heard in reality, or exists in dreams, memories, imagination, or illusions.
- **Negated (Absence of sound):** Used to mark the state of sound absence, such as “soundless” or “unheard”.
- **Extraction Method and Confidence Score:** Records the confidence score of the LLM determining whether the event is a sound event.
- **Data Validation and Cleaning:** Before the comprehensive extraction, I first made a Minimum Viable Product (MVP), extracting 20 poems for manual annotation to evaluate the LLM (DeepSeek, n.d.)’s accuracy. After the LLM ran all the data, I conducted manual checks and filtered out sound events misidentified by the LLM based on a specific Confidence Score. Ultimately, I got a database with only real sound events and high accuracy.

2.3 Geocoding

This is the most complex step in the database, which is matching the occurrence location for the sound events in the poems. I designed a priority Pipeline:

- **Highest priority: Mentioned in poems.** If the poet directly mentioned the location in the poem (such as vague small places like a small tower or an ancient road, or specific city names), the accuracy is the highest. I let the LLM re-process the sound extraction work and synchronously mark the occurrence location, and the specific city names were directly entered into the database.
- **2nd priority: Mentioned in the title or preface.** I let the LLM process the titles and prefaces of the poems to mark whether the creation location was mentioned in them.
- **3rd priority: Cross-comparison with Souyun website (Wang, n.d.).** I utilized the poetry map from the “Souyun Website” compiled by Professor Wang Zhaoqun. By writing a web crawler, I inputted the poet or poem name to match the coordinates.
- **4th priority: Cross-verification by human and LLM.** For the remaining unmatched poems, I combined the life trajectories of the poets and commentary literature, such as *Comprehensive Interpretation of Three Hundred Tang Poems* (Jin & Hengtang Tuishi, 2017) and *Comprehensive Interpretation of Three Hundred Song Ci* (Cai & Shangjiang Cunmin, 2011). I used two different LLMs (DeepSeek, n.d. and Google, n.d.) for cross-comparison to extract consensus, supplemented by my own manual verification.

When building this set of geocoding logic, I referred to the practical cases and methods of historical data geocoding in the reading material *The Programming Historian* (*The Programming Historian*, n.d.).

Up to this step, from the initially extracted 700+ sound events, I finally determined 644 pieces of address information (including certain and uncertain ones). Among them, about 30+

were based on creation locations guessed by later generations. Since they are not very certain, I marked their Certainty as “F”.

3. QGIS Spatial Mapping and Cartography

When making the map, I encountered some challenges:

3.1 Basemap Selection and Redrawing: I originally wanted to find vector maps of historical maps in CHGIS (China Historical Geographic Information System), but found that it only marked the administrative regions directly and tightly ruled by the dynasty, which was not complete enough. Ultimately, I chose the “Spatiotemporal Historical Architecture of Chinese Civilization” map (Academia Sinica, n.d.) based on “Tan Qixiang’s Edition of The Historical Atlas of China”. Since it is an uneditable raster map/image, and the Northern/Southern Song Dynasties had a chaotic situation of coexisting with multiple regimes like Liao and Jin, I manually drew the actual spheres of influence of the Tang, Northern Song, and Southern Song dynasties using QGIS, converting them into three Polygons. Here, I referred to the contents about *Working with WMS Data (QGIS3)* and *Digitizing Map Data (QGIS3)* in the QGIS tutorials (The Programming Historian, n.d.).

3.2 Symbolology:

I hoped the overall presentation would have the style of traditional Chinese landscape painting.

- **Color:** I selected five traditional colors from Chinese traditional painting to represent the five sound categories respectively.
- **Shape:** All points used “ink drop” vector graphics (SVG) made by myself. For the “Negated (soundless/absence)” state, I used hollow ink drops or solid ink drops to distinguish them.
- **Transparency:** Used to express accuracy. Points with certain locations have low transparency, and uncertain points have high transparency.
- **Dynasty distinction:** Distinguished by the boundary lines or the color of the entire Polygon.
- **Heatmap:** I also created a density heatmap of sound distribution to prominently mark the areas where sound events are most dense.
- **Basemap:** I ultimately chose Mapzen Global Terrain (Mapzen, n.d.) because it doesn’t have modern place name labels like OpenStreetMap, giving the audience a better sense of historical immersion.
- **Display:** I enabled HTML display to show the details of every single sound entry, including original phrase, sound source, context window and so on. Unfortunately, the Print Layout cannot support interactive elements.

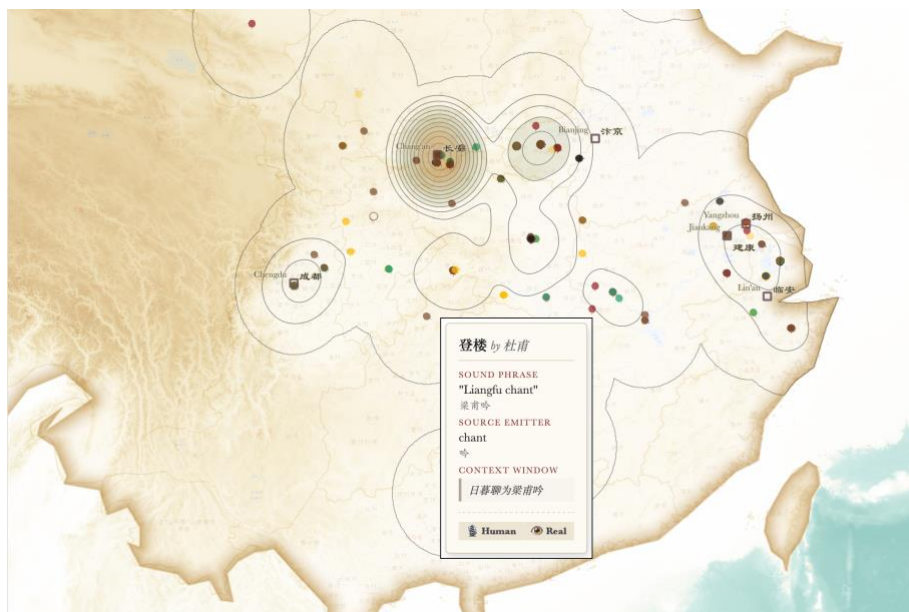


Figure 1 Display of details

3.3 Layout & Visualization

When designing the Layout in QGIS, the chapter *Becoming a Cartographer: Creating a Layout* in *The Geospatial Humanities* (MacFarlane et al., n.d.) was very useful. I finally adopted a vertical side-by-side (vertical arrangement) strategy for the maps of the three dynasties to visually contrast the “sense of vertical freefall” generated by the sound center as the political center moved south across the three dynasties. To more intuitively analyze the differences in sound categories across different dynasties, I additionally created “The Category Distribution of Tang and Song Soundscape”. (Note: I also made a comparison chart about poets’ preferences for describing sounds, but I haven’t put it into the Layout yet).

4. Key Findings

Overall Spatial-Temporal Patterns

- **Capital-Driven Shift:** The center of soundscapes shifted from Chang’an to Bianjing, and finally to the Jiangnan region (Lin’an/Jiankang). This perfectly aligns with the capital cities of the three periods, indicating that poets predominantly clustered around political centers.
- **Genre and Instruments:** Song Dynasty poetry (“Ci”) features a significantly higher proportion of instrument sounds than Tang poetry. This reflects the nature of “Ci,” which were essentially lyrics composed to be sung with musical accompaniment.
- **North vs. South:** Across all periods, southern regions consistently show more descriptions of musical instruments and natural sounds compared to the north.

Tang Dynasty

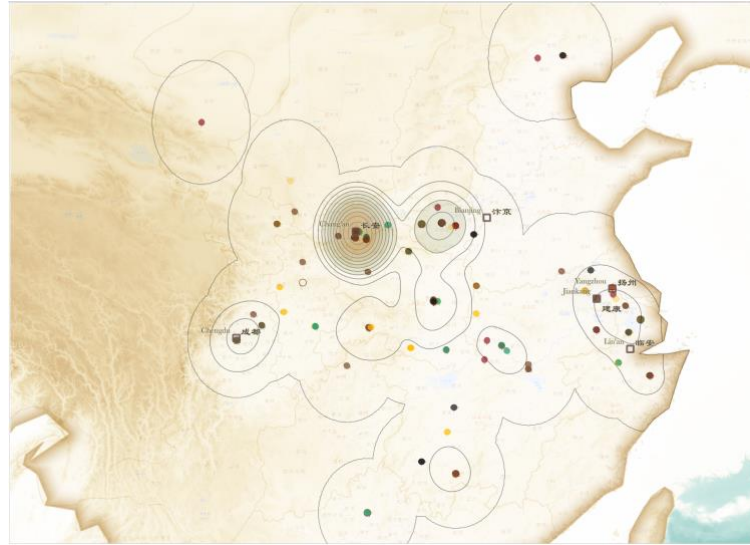


Figure 2 Details of the Tang Soundscape

- **Broadest Reach:** The primary soundscape center was the Chang'an-Luoyang area, with secondary hubs in Jiangnan and Sichuan. The geographic distribution was the widest, reaching as far northwest as Luntai.
- **Regional Variations:** Human-related sounds dominated the Chang'an-Luoyang area (followed by nature and animals), while the Jiangnan region already showed a preference for instrumental sounds.

Northern Song Dynasty



Figure 3 Details of the Northern Song Soundscape

- **Radiating from the Capital:** Most sound events radiated from the capital, Bianjing, alongside a strong presence in Jiangnan and scattered points in Chang'an and Hunan.

- **Nature in the City:** Interestingly, Bianjing’s soundscape featured many nature and animal-related sounds, whereas Jiangnan continued its strong focus on music.

Southern Song Dynasty

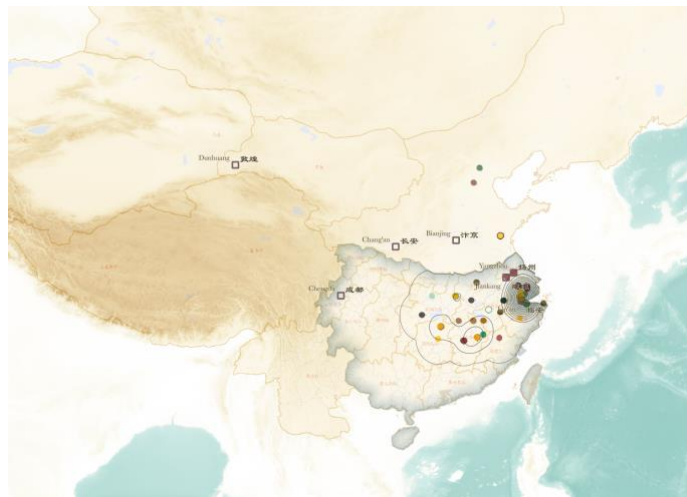


Figure 4 Details of the Southern Song Dynasty

- **Extreme Southward Shift:** The vast majority of sound events were concentrated in the Jiangnan region (centered around Lin’an) and Jiangxi.
- **Sounds Beyond Borders:** Notably, some sound events were mapped in the North, far outside the actual borders of the Southern Song (reflecting memories or longing for lost territories).
- **Fewer Human Sounds:** Compared to previous eras, there was a noticeable decrease in the description of human-related sounds.

5. Ethical Considerations and Project Reflection

5.1 Ethical Considerations: The ancient poetry involved in this project belongs to public domain content from hundreds or thousands of years ago, and the LLM APIs used in the project are all officially paid for, so the ethical risk is extremely low.

5.2 Self-Reflection

- **5.2.1 Soundscape Map or Creation Map?** This is an important thought I had after finishing the data. I found that the sounds heard highly overlap with the poets’ creation locations because poets mostly described their surrounding environments. This seems to indicate that my “soundscape map” is essentially a “poetry creation map” in terms of geographic distribution. Therefore, the true value of this project might not lie in the geographic location itself, but in looking at the differences in the categories of sounds heard in each place.
- **5.2.2 Literati Gathering, Capital Gravity, or Inclusion Bias?** I am also thinking, most of the sound events, or rather the poetry creation locations, are near the capital

circles. This means that the vast majority of the authors whose works are included in *Three Hundred Tang Poems* and *Three Hundred Song Ci* were active near the capital circles, especially towards the Southern Song dynasty where they concentrated towards the temporary capital Lin'an. Is this a coincidence? Is it that literati more actively moved closer to the capital, or does the capital circle naturally produce poets, or were only the works of literati active near the capital circle included? These still require my further research.

- **5.2.3 Future Outlook:** My initial wish was to explore different poets' preferences for sound descriptions. Although it is hard to display this multi-dimensional personal preference in QGIS, I have already tried to make it into a webpage. While this goes beyond the scope of this project, it is a huge gain for me.



Figure 5 Web Page I made

References

- Academia Sinica. (n.d.). *Zhonghua wenming shikong jichu jiagou* [Chinese civilization in time and space]. Retrieved from <https://ccts.sinica.edu.tw/>
- Cai, Y. (Annot.), & Shangjiang Cunmin (Comp.). (2011). *Song ci san bai shou quan jie* [Comprehensive interpretation of three hundred Song ci] (2nd ed.). Fudan University Press.
- chinese-poetry. (n.d.). *Chinese-poetry* [Data set]. GitHub. <https://github.com/chinese-poetry/chinese-poetry>
- DeepSeek. (n.d.). *DeepSeek* [Large language model]. <https://www.deepseek.com/>
- Gandhi, U. (n.d.). *QGIS tutorials and tips*. Retrieved from <https://www.qgistutorials.com/en/index.html>
- Google. (n.d.). *Gemini 3.1 Pro* [Large language model]. <https://gemini.google.com/>
- Jin, X. (Annot.), & Hengtang Tuishi (Comp.). (2017). *Jin Xingyao zhu Tang shi san bai shou* [Jin Xingyao's annotations on three hundred Tang poems]. Beijing United Publishing.
- MacFarlane, J., et al. (n.d.). *Geospatial humanities*. UPEI Pressbooks. Retrieved from <https://pressbooks.library.upei.ca/geospatialhumanities/>
- Mapzen. (n.d.). *Mapzen global terrain* [Data set]. Registry of Open Data on AWS. <https://registry.opendata.aws/terrain-tiles/>
- The Programming Historian: Mapping*. (n.d.). Retrieved from <https://programminghistorian.org/en/lessons/?topic=mapping>
- Wang, Z. (n.d.). *Tang Song wenxue biannian ditu* [Chronological map of Tang and Song literature]. Souyun. Retrieved from <https://cnkgraph.com/Map/PoetLife>